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SynopStudio - AI Tool for Automatic Video Summarization and Editing

Software Engineering Project

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MileStone 1

Introduction

1.1 Context

Video consumption on social media is increasingly concentrated in short, vertical formats such as Instagram Reels, YouTube Shorts, and TikTok. To reach these publics, content creators, whether hobbyists or professional communication teams, must turn a long recording into clips of a few seconds. This involves several manual steps: watching the full recording, identifying relevant moments, cutting, captioning, reframing the image for each platform, and writing titles and descriptions. The process is time consuming and requires video editing skills.

Recent advances in artificial intelligence have made it possible to automate part of this work. Speech recognition models produce transcripts with word-level timestamps and can distinguish between speakers, and large language models can identify topics, summarize content, and suggest excerpts.

This automation benefits a lot of users. For hobbyists, such as streamers, podcasters and people who record talks, tutorials or personal projects, it lowers the barrier to publishing because they can produce short clips without editing skills or expensive software. For institutions such as universities, research centers and civil society organizations, it makes it viable to disseminate the large volume of lectures, conferences and debates they produce, most of which never reaches an audience beyond those who watched it live.

However, automating editorial decisions raises its own challenges. An automatically selected excerpt can remove the context a statement depends on, and automatically generated titles or descriptions can attribute to speakers things they never said. The problem is most serious for informative content, such as academic, scientific or political material, but it matters to any creator who wants the published clip to reflect what was actually said. Users therefore need to understand why each excerpt was chosen, be able to check it against the original recording, and have the final say over what is published.

This project addresses these needs by developing a web application that turns long videos into short clips ready for social media. The application automates the most time consuming steps of editing, keeps every clip traceable to the original recording, and leaves the final decision to the user.

1.2 Problem

The conversion of long form video into short, shareable clips remains a slow and largely manual process. Reviewing and trimming hours of footage to find the moments that matter is an overwhelming task in itself, and the work does not end once a segment is selected. Each clip must then be reformatted to match the aspect ratios and length limits of different platforms, such as vertical 9:16 for short form feeds versus 16:9 for traditional video players. This requires dedicated editing software, and, in many cases, qualified editors hired from outside the team. As a result, production costs rise and turnaround times grow, since every clip depends on coordination with third parties.

Captions add a second layer of effort. The manual creation of synchronized, timestamped captions and their translation into other languages is highly labor intensive. Automatic speech recognition reduces part of this workload, but general purpose captioning tools frequently mishandle domain specific terminology. This makes them unreliable for technical or academic content, where a single misrecognized term can change the meaning of an entire sentence.

Speaker identification is a related challenge. It is particularly difficult to distinguish speakers by voice alone when several participants have similar vocal profiles. When automatic methods fail, creators must scrub through the footage and label each speaker by hand.

Finally, current approaches to AI generated clips typically process the full video with only a basic prompt as guidance. Without supporting material such as presentation slides or related articles, the model lacks the context it needs to judge which segments are genuinely relevant. The resulting clips are often generic or miss the key points of the content.

1.3 Goals and Expected Results

The platform aims to leverage AI trained on existing short-form clips from diverse sources to efficiently identify relevant segments and speakers within long-form videos. It will support context file integration to enhance transcription accuracy and terminology interpretation, while providing synchronized transcriptions that link phrases to their original instances and distinguish speakers. Content will be segmented by topic, enabling natural language search and selection, and the AI will propose clips with configurable durations and clear structure, prioritizing relevance and clarity. An editor will allow video review and editing directly from the synchronized transcription, all built on a modular architecture featuring a documented API, asynchronous processing, and open LLM modules.

2. State of the Art

2.1 Related Work

2.1.1 OpusClip

OpusClip is an AI tool designed primarily for content creators, marketers, and podcasters to transform long videos into viral short clips. The platform's main objective is to automate the process of extracting engaging moments by analyzing video content, generating captions, and reframing speakers for vertical video formats. However, OpusClip focuses heavily on social media metrics and viral potential rather than informational completeness or scientific rigor. OpusClip's key features include automatic detection of highlights on long videos, generating edited short videos based on highlights focused on social media engagement, automated synchronized captioning and reframing, and video adaptation for multiple platforms.

2.1.2 CapCut

CapCut is a multi-platform video editor developed by ByteDance. It is widely used by social media content creators as it provides a simple and intuitive interface for video production. The platform also provides visual effects, filters, timeline editing tools, and AI-assisted features. While it is highly versatile for general video editing, it lacks features such as automated editing based on user specifications and the integration with academic or institutional reference sources. CapCut's main features include professional visual effects, animations, transitions, audio editing, automated subtitling based on speech and collections of pre-made templates.

2.1.3 Canva

Canva is an online platform focused on graphic design and visual communication. Canva has expanded its core features and now includes basic video editing and content generation using AI tools. The main purpose is to be an easy alternative to support users with little design experience in creating presentations and simple edited videos. Nevertheless, this platform is not intended for video summarization and creation of highly structured videos. Canva's key features include an intuitive interface based on drag and drop, support for multiple users to collaborate on the same project, basic video editing tools such as trimming videos, apply animations, subtitles and background music, image generation and several presentation templates that can be edited through the AI editor.

